

Mohsen Fayyaz

CS PH.D. STUDENT · UNIVERSITY OF CALIFORNIA, LOS ANGELES

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Education

University of California, Los Angeles

PH.D. IN COMPUTER SCIENCE

Los Angeles, CA

Sep. 2023 - Present

- advisor: **Nanyun (Violet) Peng**
- GPA: 3.95/4

University of Tehran

MASTER'S IN ARTIFICIAL INTELLIGENCE, COMPUTER ENGINEERING

Tehran, Iran

Sep. 2021 - Aug. 2023

- advisor: Yadollah Yaghoobzadeh
- GPA: 20/20 (4/4)

University of Tehran

B.SC. IN COMPUTER ENGINEERING

Tehran, Iran

Sep. 2017 - Sep. 2021

- Ranked **3rd** among 103 Computer Engineering students
- GPA: 19.08/20

Research Interests

- Natural Language Processing
- Deep Learning

Publications

11. Multilingual Routing in Mixture-of-Experts

ICLR 2026

- L Bandarkar, C Yang, **M Fayyaz**, J Hu, N Peng

10. Steering MoE LLMs via Expert (De)Activation

ICLR 2026

- **M Fayyaz**, A Modarressi, H Deilamsalehy, F Dernoncourt, R Rossi, T Bui, H Schütze, N Peng

9. Collapse of Dense Retrievers: Short, Early, and Literal Biases Outranking Factual Evidence

ACL 2025

- **M Fayyaz**, A Modarressi, H Schuetze, N Peng

8. MRAG-Bench: Vision-Centric Evaluation for Retrieval-Augmented Multimodal Models

ICLR 2025

- W Hu, J Gu, Z Dou, **M Fayyaz**, P Lu, K Chang, N Peng

7. Evaluating human alignment and model faithfulness of LLM rationale

arXiv preprint

- **M Fayyaz**, F Yin, J Sun, N Peng

6. SkipPLUS: Skip the First Few Layers to Better Explain Vision Transformers

CVPR Workshops 2024

- F Mehri, **M Fayyaz**, MS Baghshah, MT Pilehvar

5. DecompX: Explaining Transformers Decisions by Propagating Token Decomposition

ACL 2023

- A Modarressi*, **M Fayyaz***, E Aghazadeh, Y Yaghoobzadeh, MT Pilehvar
- (* equal contribution)

4. BERT on a Data Diet: Finding Important Examples by Gradient-Based Pruning

ENLSP @ NeurIPS 2022

- **M Fayyaz***, E Aghazadeh*, A Modarressi*, MT Pilehvar, Y Yaghoobzadeh, SE Kahou
- (* equal contribution)

3. GlobEnc: Quantifying Global Token Attribution by Incorporating the Whole Encoder Layer in Transformers

NAACL 2022

- A Modarressi*, **M Fayyaz***, Y Yaghoobzadeh, MT Pilehvar
- (* equal contribution)

2. Metaphors in Pre-Trained Language Models: Probing and Generalization Across Datasets and Languages

ACL 2022

- E Aghazadeh*, **M Fayyaz***, Y Yaghoobzadeh
- (* equal contribution)

1. Not All Models Localize Linguistic Knowledge in the Same Place: A Layer-wise Probing on BERToids' Representations

BlackboxNLP @ EMNLP 2021

- M Fayyaz*, E Aghazadeh*, A Modarressi, H Mohebbi, MT Pilehvar
- (* equal contribution)

Work Experience

Research Intern

ADOBE

- Steering MoE LLMs via Expert (De)Activation

Seattle, Washington, United States

Jun. 2025 - Sep. 2025

Data Scientist

DIVAR

- Developed NLP services including various ML tasks and models at large scale.

Tehran, Iran

Jul. 2023 - Aug. 2023

Data Scientist

SOTOON

- Developed NLP/AutoML services including various ML tasks and models at large scale.

Tehran, Iran

Jul. 2022 - Jun. 2023

Machine Learning Engineer

TAPSELL

- Developed NLP engines and ML models for enhancing targeted advertising.

Tehran, Iran

Sep. 2021 - Jul. 2022

Research Experience

Graduate Student Researcher at University of California, Los Angeles

UNDER SUPERVISION OF PROF. NANYUN (VIOLET) PENG

- Ph.D. Researcher at PLUS Lab

Los Angeles, CA

Fall 2023 - Present

Remote Collaborator at École de technologie supérieure

UNDER SUPERVISION OF PROF. SAMIRA EBRAHIMI AND PROF. YADOLLAH YAGHOOBZADEH

- BERT on a data diet: Finding important examples by gradient-based pruning (ENLSP @ NeurIPS 2022)

Montreal, Canada

Spring 2022

Graduate Research Assistant at University of Tehran

UNDER SUPERVISION OF PROF. YADOLLAH YAGHOOBZADEH AND PROF. MOHAMMAD TAHER PILEHVAR

- Quantifying global token attribution by incorporating the whole encoder layer in Transformers. (NAACL 2022)
- Metaphors in pre-trained language models: Probing and generalization across datasets and languages. (ACL 2022)

Tehran, Iran

2021 - 2023

NLP Research Intern at Tehran Institute for Advanced Studies (TeIAS)

UNDER SUPERVISION OF PROF. MOHAMMAD TAHER PILEHVAR

- Layer-wise probing on BERT, XLNet, and ELECTRA representations for linguistic knowledge. (BlackboxNLP @ EMNLP2021)

Tehran, Iran

Summer 2021

Undergraduate Research Assistant at University of Tehran

UNDER SUPERVISION OF PROF. AZADEH SHAKERY

- Investigating toxicity detection knowledge in contextual language models and possible biases.

Tehran, Iran

Spring 2021

Honors & Awards

- 2021 **Enrolled in the graduate school as a top student**, Without passing the entrance examinations
- 2020 **F.O.E (Faculty of Engineering) Award**, Ranked 1st among all of 103 Computer Engineering students
- 2019 **F.O.E (Faculty of Engineering) Award**, Ranked 3rd among all of 103 Computer Engineering students
- 2017 **University of Tehran Scholarship**, Received scholarship from the University of Tehran Sponsors Foundation as an exceptional talent
- 2017 **Ranked in top 0.4% in Konkur Exam**, Of all 148,429 participants in Iran
- 2015 **Second place in Khwarizmi Youth Award**, An important national research award given annually by the President of Iran.

University of Tehran

University of Tehran

University of Tehran

University of Tehran

Iran

Iran

Teaching

TEACHING ASSISTANTSHIP

2025	CS162 - Natural Language Processing , Prof. Nanyun (Violet) Peng	UCLA
2024	CS263 - Natural Language Processing , Prof. Nanyun (Violet) Peng	UCLA
2020-2022	Artificial Intelligence , Prof. Yaghoobzadeh, Dr. Fadaei	University of Tehran
2019-2020	Formal Languages and Automata Theory , Prof. Hojjat	University of Tehran
2019-2020	Advanced Programming , Prof. Khosravi	University of Tehran
2019	Digital Logic Design , Prof. Navabi	University of Tehran

Professional Development

The following online courses were taken to acquire skills relevant to my research:

- Build Basic Generative Adversarial Networks (GANs), Coursera.org - Feb 2021 - Credential
- Deep Learning Specialization, Coursera.org - Sep 2020 - Credential
- Sequence Models, Coursera.org - Sep 2020 - Credential
- Convolutional Neural Networks, Coursera.org - Aug 2020 - Credential
- Structuring Machine Learning Projects, Coursera.org - Aug 2020 - Credential
- Improving Deep Neural Networks: Coursera.org - Aug 2020 - Credential
- Neural Networks and Deep Learning, Coursera.org - Aug 2020 - Credential

Skills

Programming	Python (PyTorch, sklearn, NumPy, Pandas, Tensorflow, PySpark), C++, Java, Java-script, SQL
Tools and Frameworks	PyTorch, Keras, Tensorflow, HuggingFace, Transformers, Git, React-Native, Android Studio
Languages	• Persian: Native • English: Fluent (TOEFL score: 116/120, Reading: 29, Listening: 30, Speaking: 28, Writing: 29)

References

Nanyun (Violet) Peng , violetpeng@cs.ucla.edu	UCLA
Mohammad Taher Pilehvar , mp792@cam.ac.uk	Cardiff University / TelAS
Yadollah Yaghoobzadeh , y.yaghoobzadeh@ut.ac.ir	University of Tehran
Hossein Hojjat , hojjat@ut.ac.ir	University of Tehran